MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MODULATOR, SYNCHRONOUS, ANALOG , KEY CODE KAD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic synchronous modulator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.1717 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KAD.

3 5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 20 required and 20 optional contact pins positioned A21 through A40 for required and A1 through A20 for optional.

3.5.5 Fin position. The module shall have an insertion and extraction on increment A and heat dissipation on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/220-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

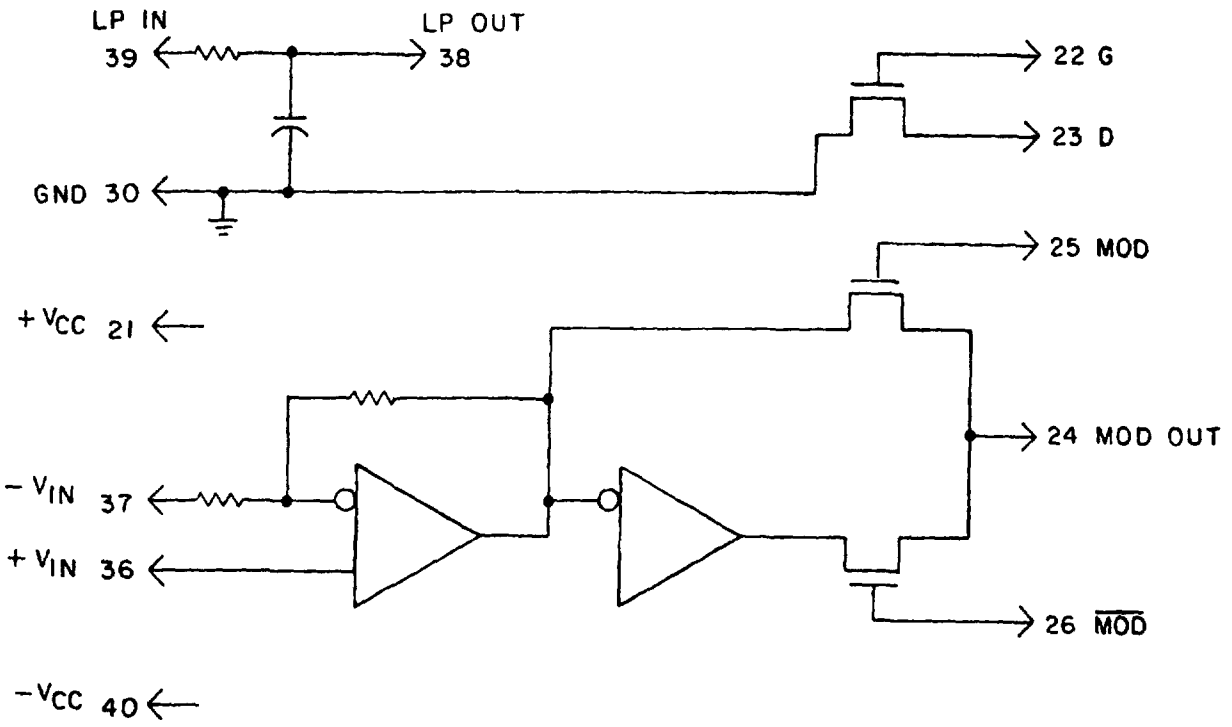
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4 4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES
1. V_{IN} = Analog input
2. G = Gate
3. D = Drain-source
4. MOD = Modulator
5. The following pins are unused - 1 through 20, 27, 28, 29, 31 through 35.

FIGURE 1. KAD module function diagram, synchronous modulator

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Units
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation	II	S	10	∞	10	∞	10	∞	10	∞	MΩ
B	Stress duration	III	S	1	---	1	---	1	---	1	---	Minute
C1-C4	Gain accuracy check (100 Hz)	IV	X	0.9	1.1	0.9	1.1	0.9	1.1	0.9	1.1	Vrms
D1-D4	Gain accuracy check (10000 Hz)	V	X	0.9	1.1	0.9	1.1	0.9	1.1	0.9	1.1	Vrms
E1,E2	Modulator response time	VI	X	0	5	0	5	0	6	0	6	µs
F1-F3	FET threshold (OFF) current	VII	X	0	10	0	10	0	10	0	10	µA
G1	Input resistance check ($+V_{IN}$) 300 kΩ minimum	VIII	S	3.70	5.05	3.70	5.05	3.65	5.10	3.65	5.10	V
G2	Input resistance check ($-V_{IN}$)	VIII	S	1.78	1.82	1.78	1.82	1.73	1.87	1.73	1.87	V
H1	Output resistance check, 400 Ω maximum	IX	S	3.53	5.05	3.53	5.05	3.48	5.05	3.48	5.05	V
H2	Output resistance check, 600 Ω maximum	IX	S	-2.75	-5.05	-2.75	-5.05	-2.70	-5.05	-2.70	-5.05	V
I	Low pass filter network (60 Hz rejection) check	X	X	-18.5	-22.0	-18.5	-22.0	-17	-23	-17	-23	dB
K1	Output offset voltage	XI	X	-15	+15	-15	+15	-20	+20	-20	+20	mV
K2	Output offset voltage	XI	X	-30	+30	-30	+30	-40	+40	-40	+40	mV
L1	FET ON resistance check (V_{IN} = -10V), 620 Ω	XII	X	0	-580	0	-580	0	-580	0	-580	mV
L2	FET ON resistance check (V_{IN} = +10 V), 205 Ω	XII	X	0	200	0	200	0	200	0	200	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Units
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1,M2	Power supply current	XIII	Q	1	8	1	8	1	8	1	8	mA
N1-N3	Gate input capacitance	XIV	Q	0	15	0	15	0	15	0	15	pF
P1-P3	Gate output capacitance	XV	Q	0	12	0	12	0	12	0	12	pF
O1,O2	Power supply transient rise time	XVI	S	25	75	25	75	25	75	25	75	µs

1/ "X" parameters shall be 100 percent tested during acceptance testing.

2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only

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4.4.2 Order of testing. Test conditions A through P shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through P in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through P shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through P shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/220 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833705. Part number M28787/220-1 supersedes part numbers 2833705 and M28787/220.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Power dissipation	600 mW maximum
Supply voltage	±15 V dc ±10 percent
Supply current	8 mA maximum
Input signal	5 Vrms maximum
Amplifier gain	Unity
Bandwidth, analog input	DC to 10 kHz minimum

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Response, modulator	5 µs maximum
Resistance	
Input	$+V_{IN}$ 300 k ohms minimum $-V_{IN}$ 56 2 k ohms ±1 percent
Output	800 ohms maximum (-5 V output) 400 ohms maximum (+5 V output)
FET switch	
Gate voltage	+7 V to -20 V maximum
OFF resistance	10 megohms
Gate threshold	-1.0 volt gate to source
Gate input capacitance	15 pF maximum
Low pass section	17 dB minimum rejection at 60 Hz

6.5 Module function. The module consists of two unity gain operational amplifie connected to feed a complementary FET switch chopper with positive and negative polarities of an input signal. An optional RC low pass filter network is included.

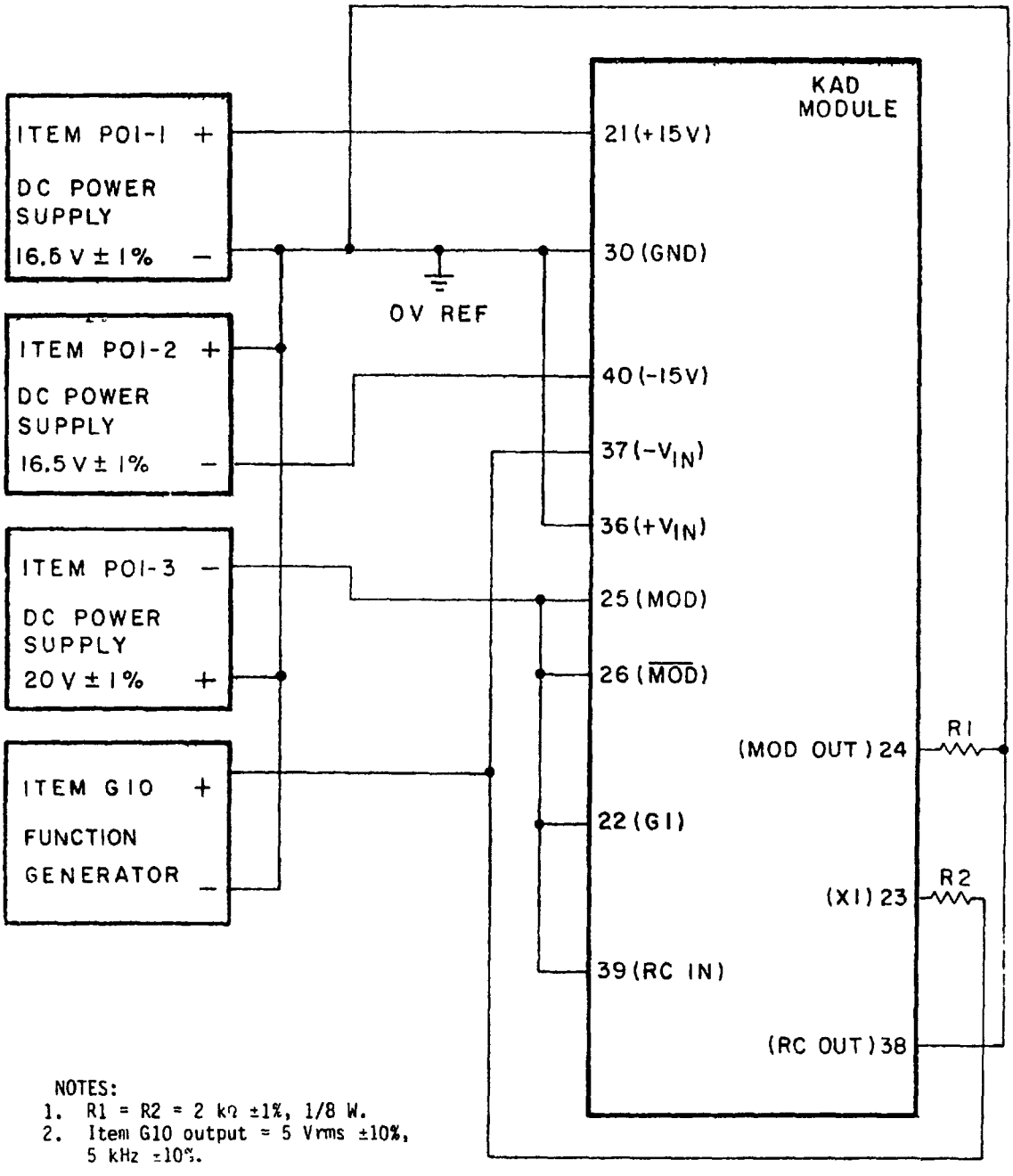
6 6 Changes from previous issue. Astericks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of th changes

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NOTES:
1. R1 = R2 = 2 kΩ ±1%, 1/8 W.
2. Item G10 output = 5 Vrms ±10%, 5 kHz ±10%.

FIGURE 2. Operating temperature and life test circuit

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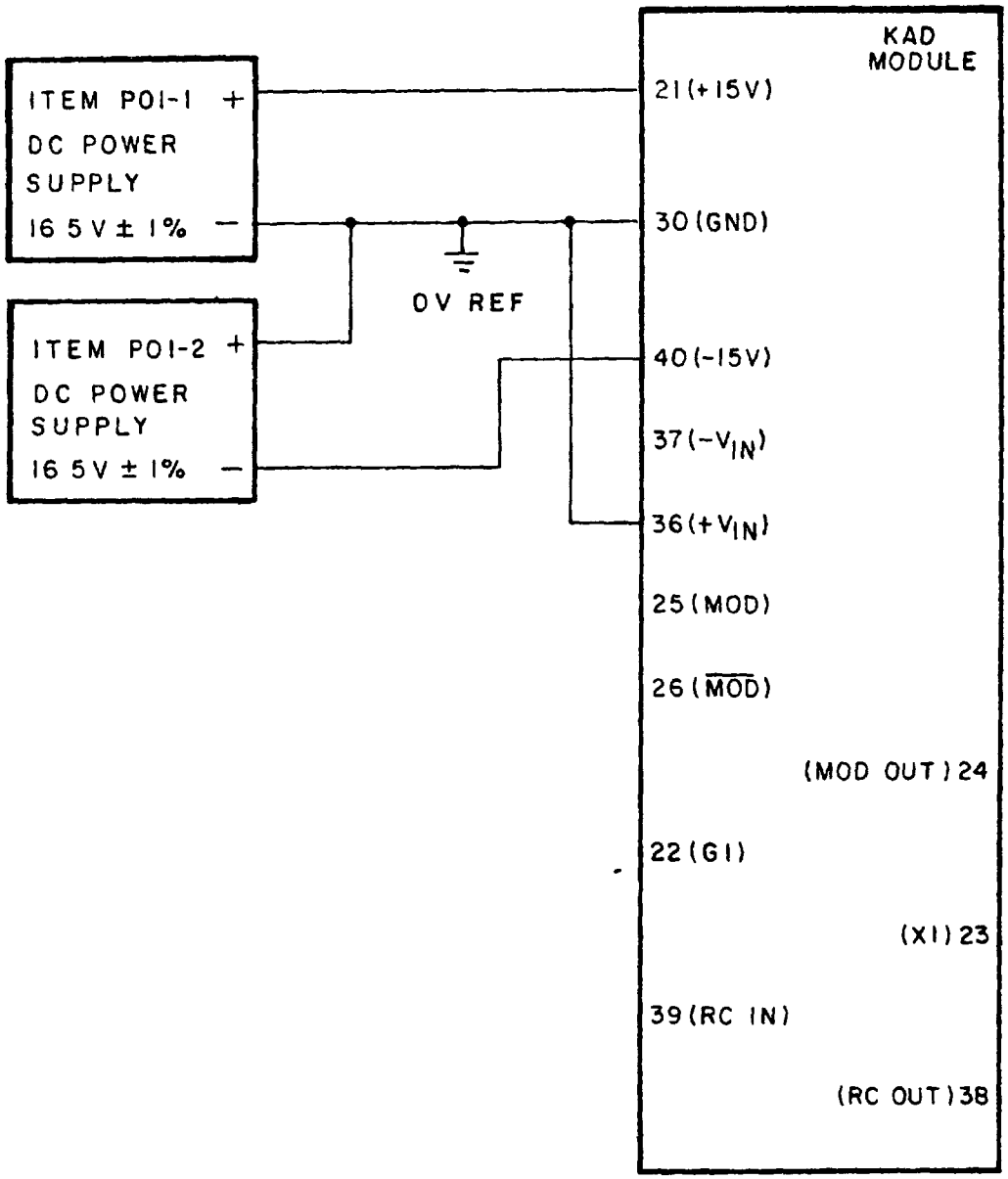


FIGURE 3. Component temperature test circuit

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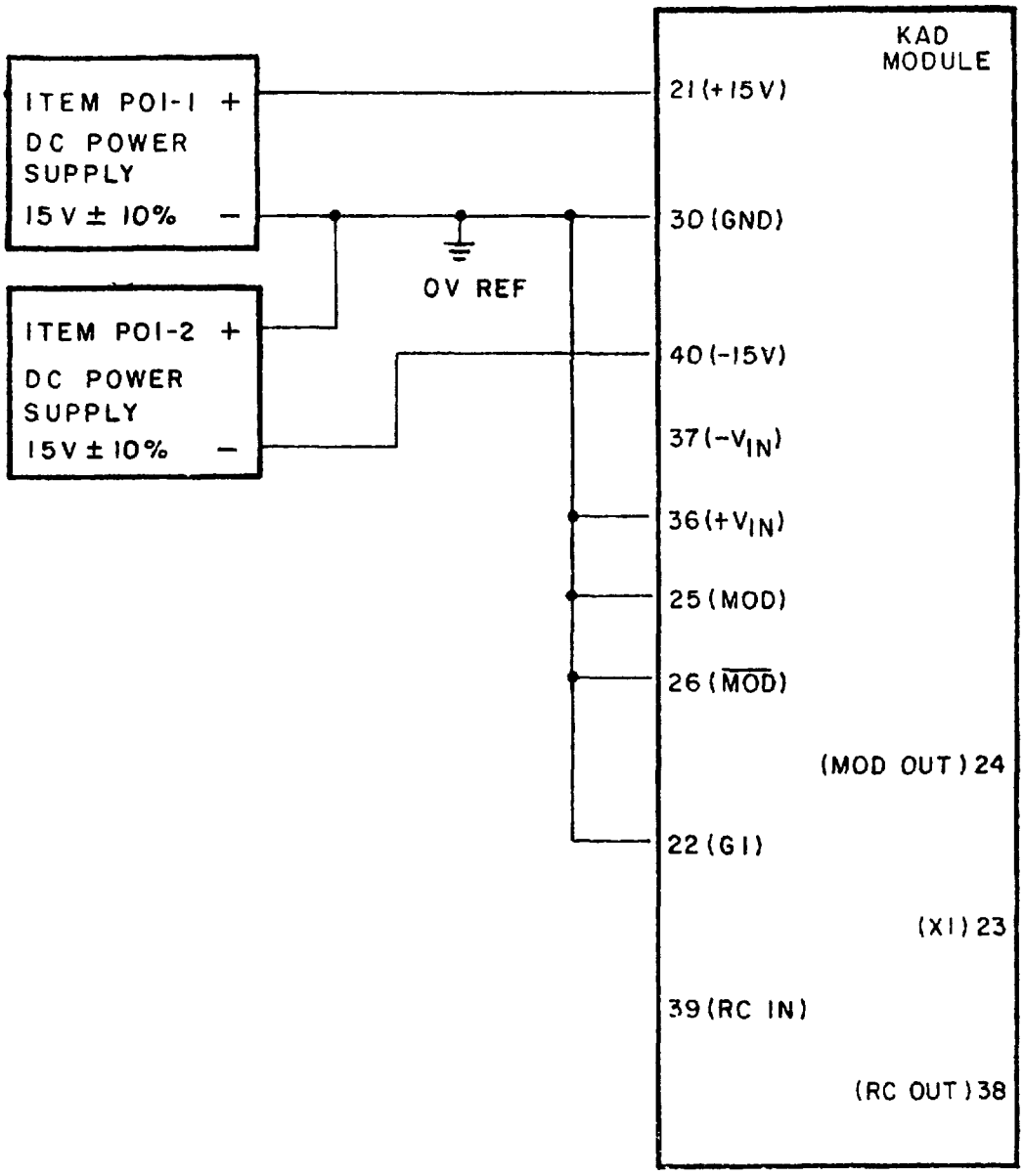


FIGURE 4 Vibration test circuit

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	All pins	Frame	10	∞	MΩ	5
A2	Isolation	21,22,23,24,25, 26,30,36,37,38, 39,40	27,28,29,31,32, 33,34,35	10	∞	MΩ	5

TABLE III. Stress duration test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
B	Stress duration	1	---	Minute	6,7

TABLE IV. Gain accuracy check (100 Hz) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Gain accuracy check (100 Hz)	36	---	0.9	1.1	Vrms	8
C2	↓	36	---	↓	↓	↓	↓
C3		37	36				
C4		37	36				

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TABLE V. Gain accuracy check (10,000 Hz) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Gain accuracy check (10,000 Hz)	36	---	0.9	1.1	Vrms	8
D2	↓	36	---	↓	↓	↓	↓
D3	↓	37	36	↓	↓	↓	↓
D4	↓	37	36	↓	↓	↓	↓

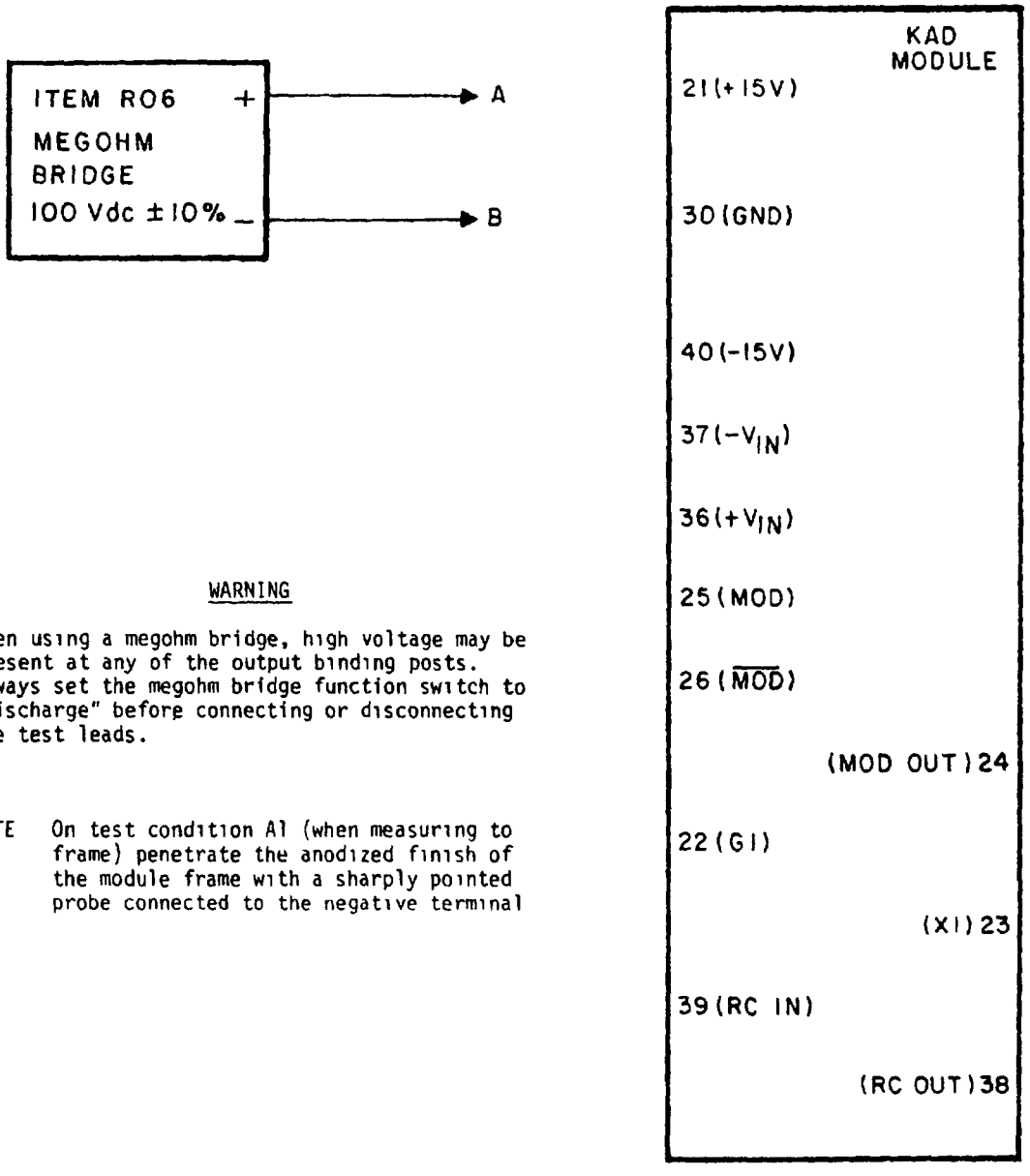
TABLE VI. Modulator response time test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to point	Min	Max		
E1	Modulator response time (T_D+)	D	0	5	µs	9,10
E2	Modulator response time (T_D-)	E	0	5	µs	9,10

TABLE VII. FET Threshold (OFF) current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	FET threshold (OFF) current	---	36,25	26,24	0	10	µA	11
F2	FET threshold (OFF) current	36	37,26	25,24	0	10	µA	11
F3	FET threshold (OFF) current	36	Open	22,23	0	10	µA	11

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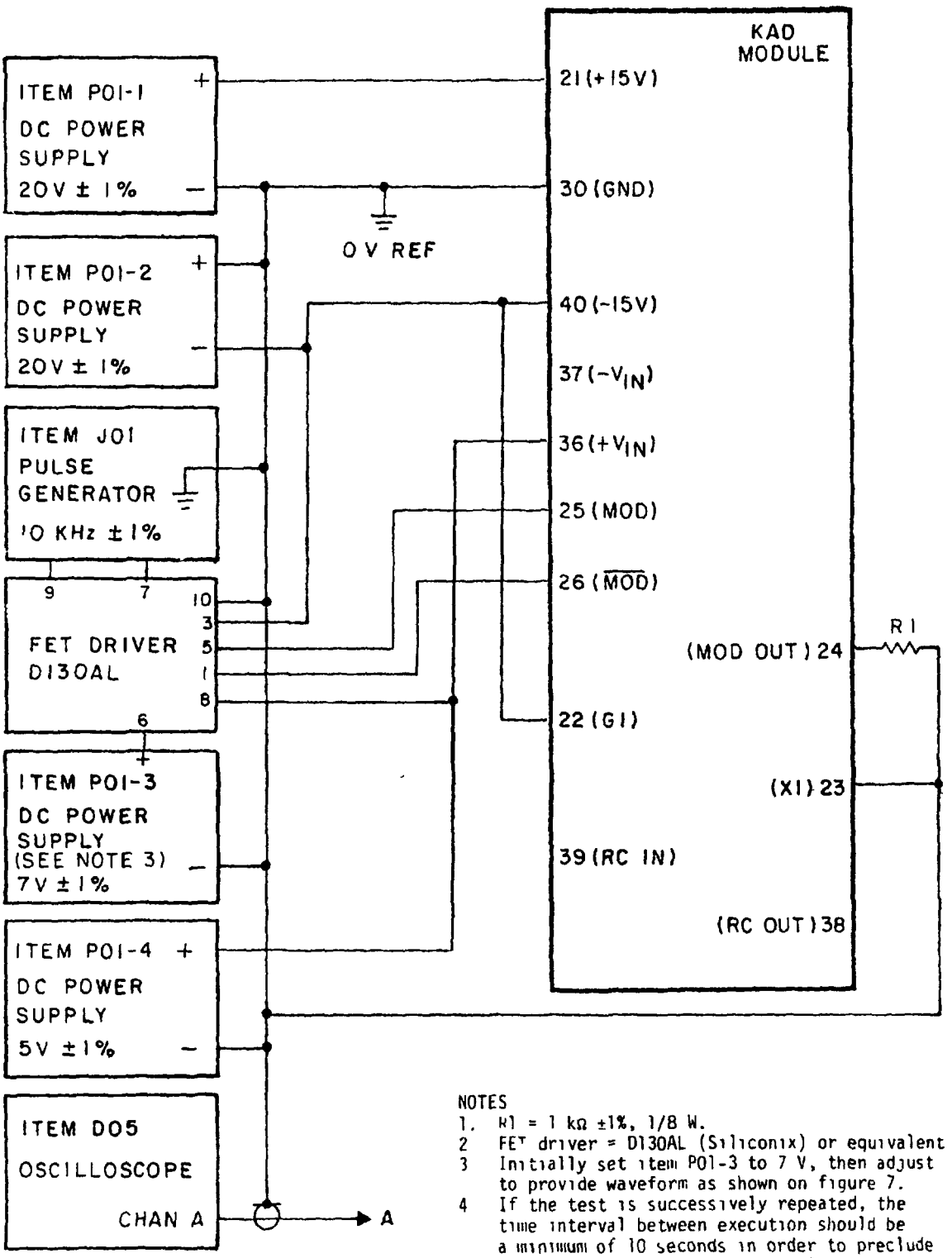
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE On test condition A1 (when measuring to frame) penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 5 Isolation test circuit for table II.

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NOTES
1. R1 = 1 kΩ ±1%, 1/8 W.
2 FET driver = D130AL (Siliconix) or equivalent
3 Initially set item P01-3 to 7 V, then adjust to provide waveform as shown on figure 7.
4 If the test is successively repeated, the time interval between execution should be a minimum of 10 seconds in order to preclude detrimental effects due to module heating

FIGURE 6 Stress duration test circuit for table III.

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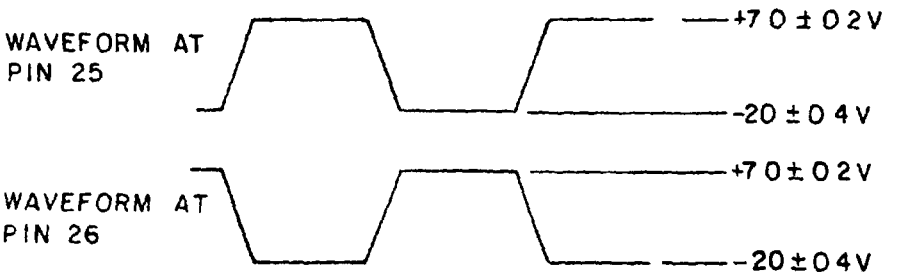
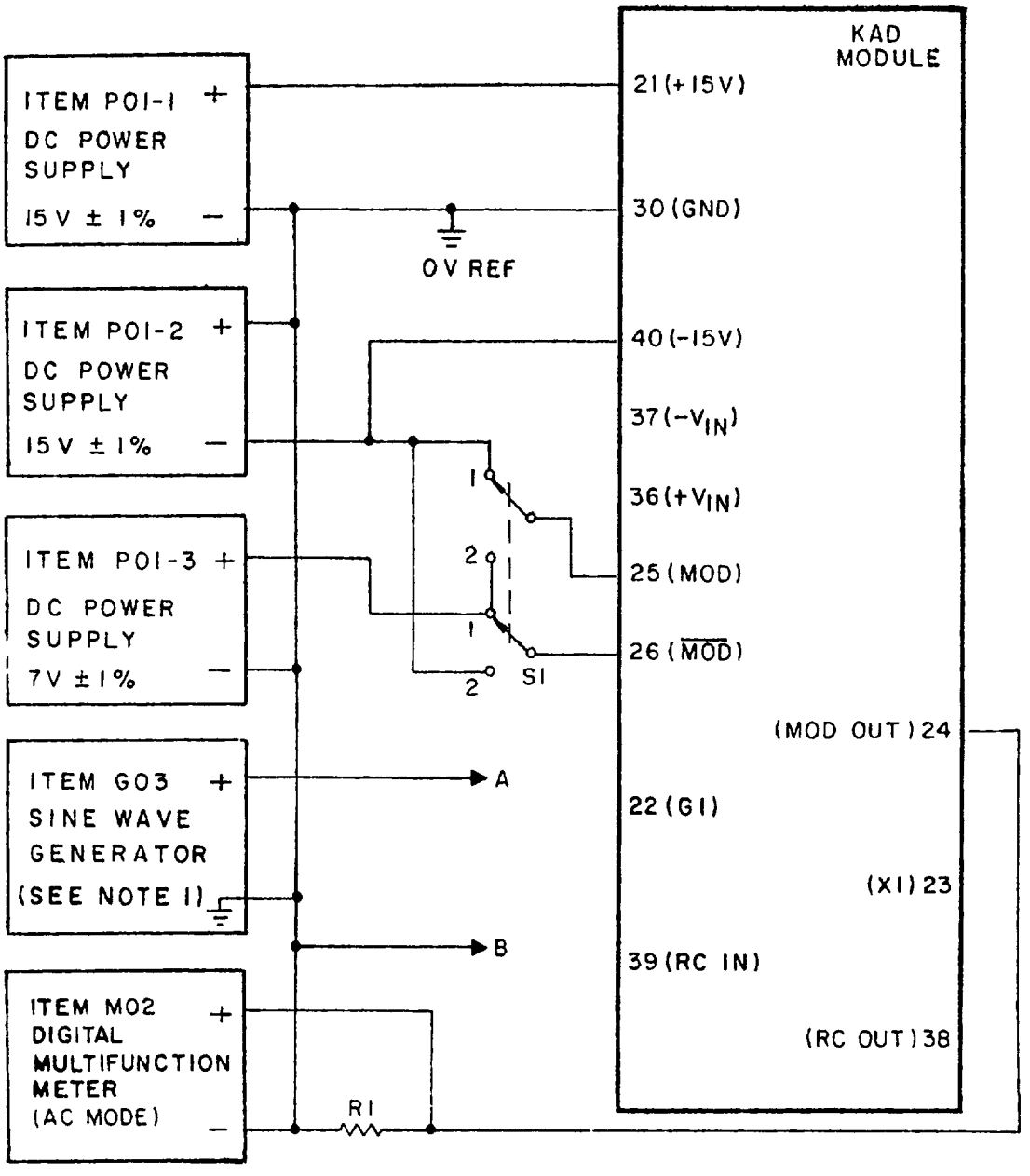


FIGURE 7 Stress duration test circuit waveforms for table II.

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NOTES
1 R1 = 10 kΩ ±1%, 1/8 W
2 S1 = DPDT switch
3 Item G03 setup.
100 ±5 Hz at 1 Vrms ±1% for test conditions C1-C4.
10, 000 ±500 Hz at 1 Vrms ±1% for test condition D1-D4.
4 Set switch S1 to position 1 for test conditions C1, C4, D1 and D4 and position 2 for test conditions C2, C3, D2 and D3

FIGURE 8 Gain accuracy check (100 and 10,000 Hz) test circuit for tables IV and V

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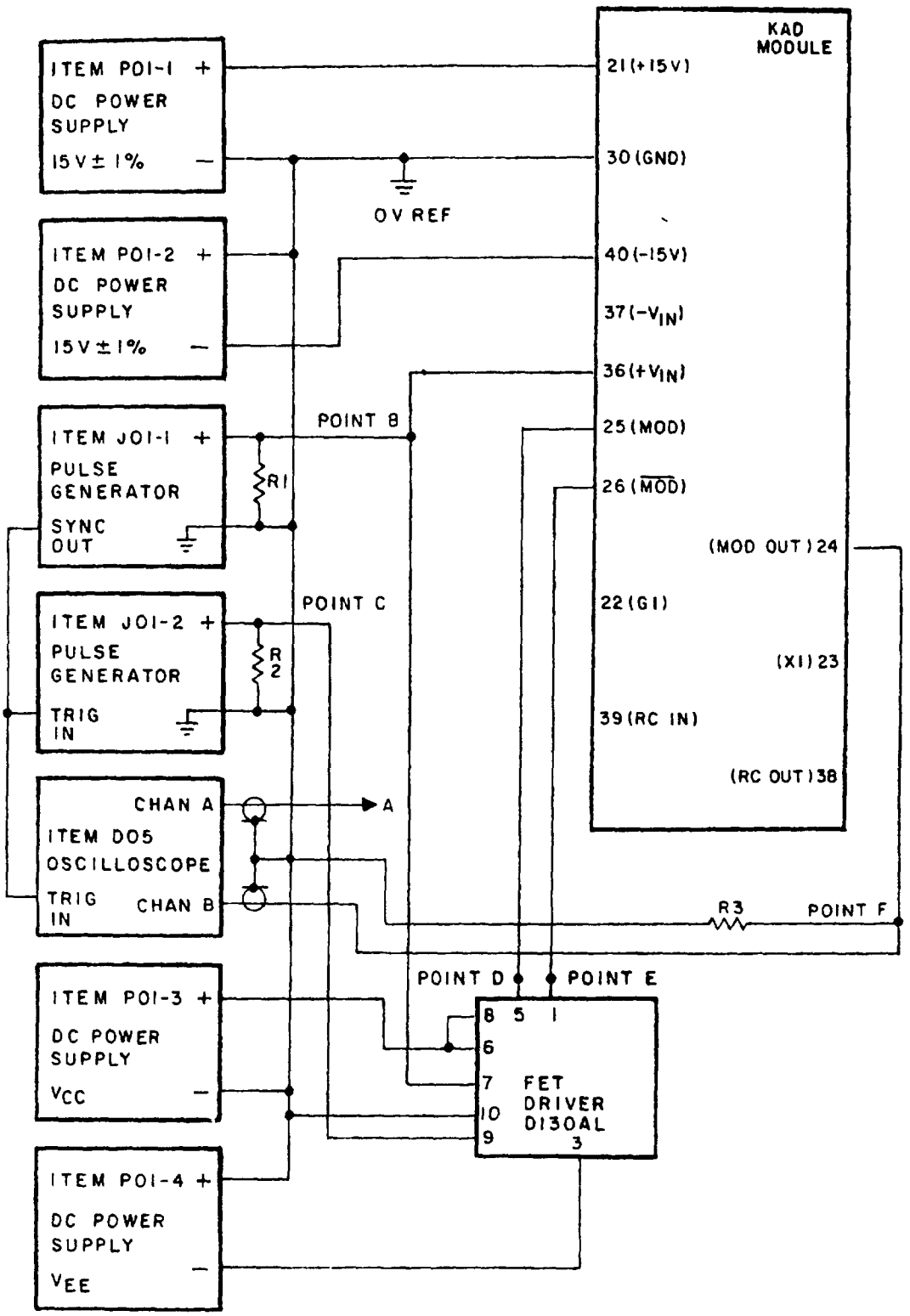
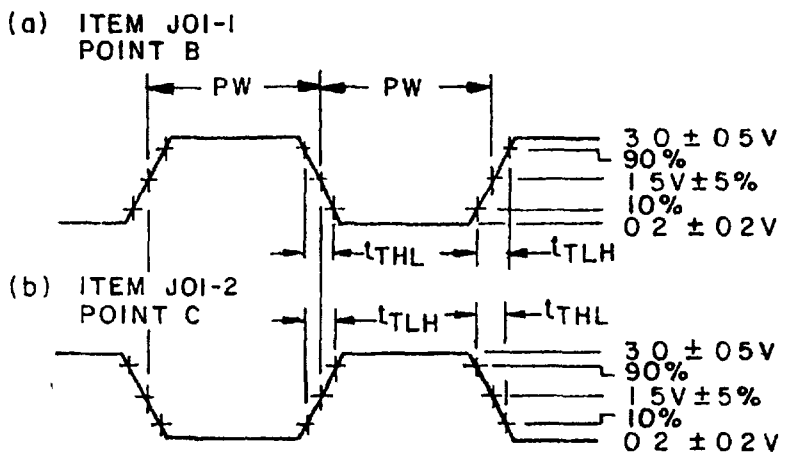


FIGURE 9 Modulator response time test circuit for table VI

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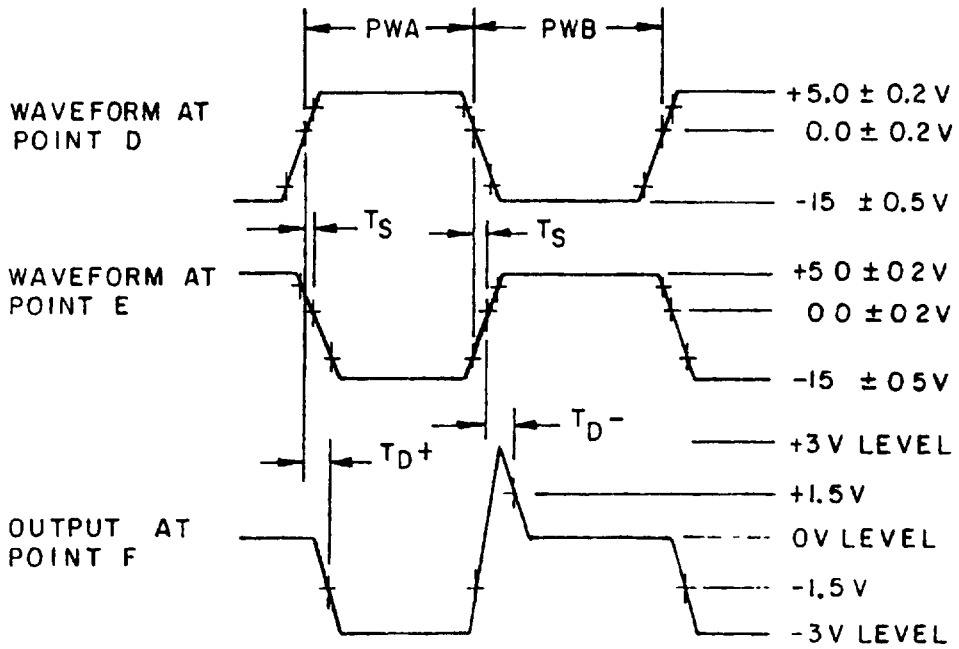
NOTES
1 R1 = R2 = 50Ω ±5%, 1 W.
2 R3 = 10 kΩ ±1%, 1/8 W.
3 FET driver = D130AL (Siliconix) or equivalent.
4 Initially adjust items J01-1 and J01-2 as follows



5 Input waveform parameters:
PW = 50 ±5 µs
$t_{TLH} = t_{THL}$ = 100 ns maximum
6 Initially adjust item P01-3 (V_{CC}) to 7 V and item P01-4 (V_{EE}) to -17 V. Adjust item P01-3 to obtain a +5 ±0.2 V level and item P01-4 to obtain a -15 ±0.5 V level as specified in waveforms at point D and E.
7 Adjust item J01-2 pulse width period and delay to obtain the specified delay between waveforms D and E.
8 Connect item D05, channel A to point D for T_D+ measurement and to point E for T_D- measurements.

FIGURE 9 Modulator response time test circuit for table VI - Continued

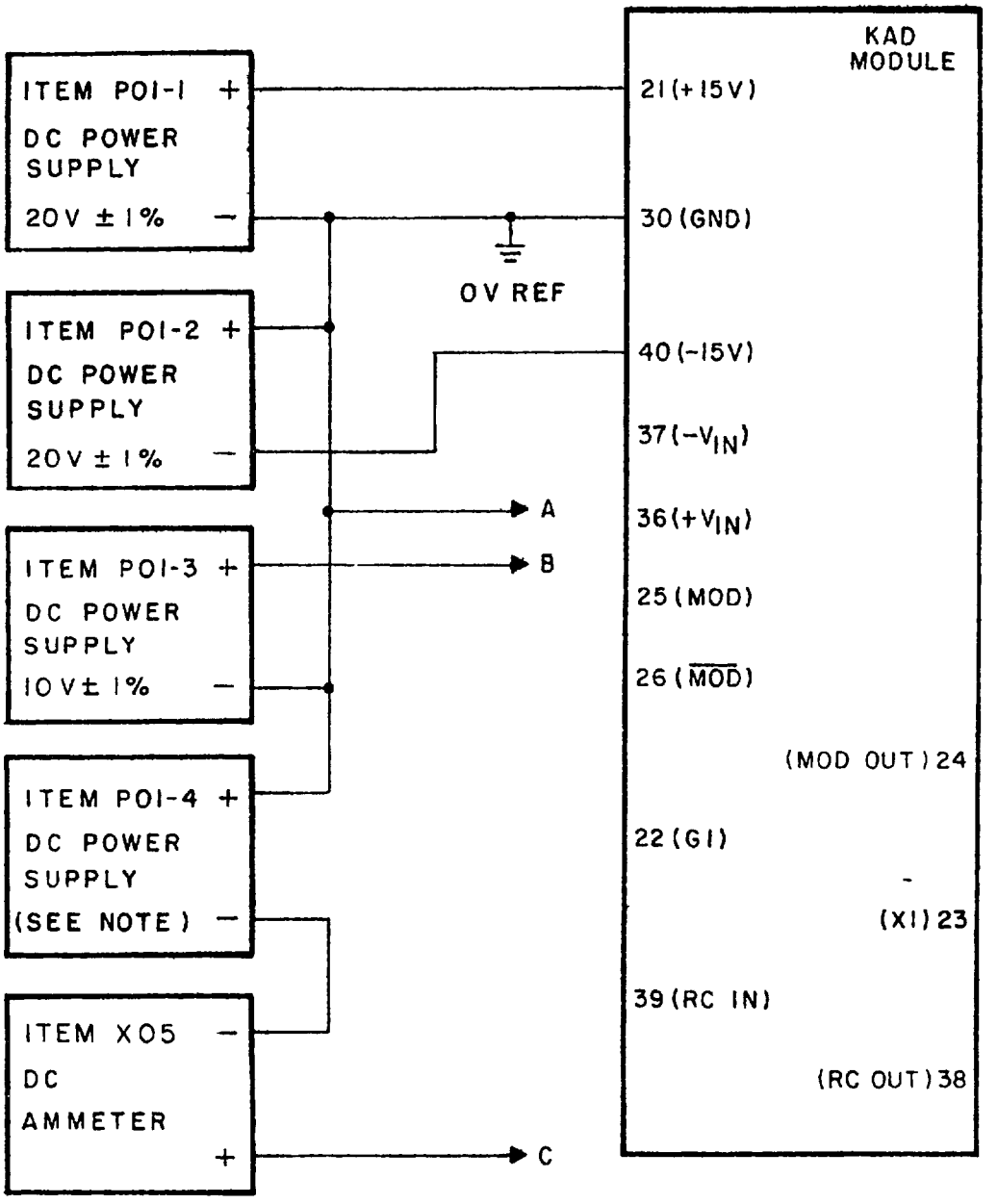
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NOTE Input pulse parameters
PWA = PWB = 50 ±5 µs
T_S = 0 ±35 ns.

FIGURE 10. Modulator response time test circuit waveforms for table VI.

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NOTE Set item POI-4 to 11 V ±1% for test conditions F1 and F2 and 1 V ±1% for test condition F3.

FIGURE 11. FEI threshold (off current) test circuit for table VII

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TABLE VIII. Input resistance check test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	Input resistance check ($+V_{IN}$), 300 kΩ minimum	36	---	3.70	5.05	V	12
G2	Input resistance check ($-V_{IN}$)	37	36	1.78	1.82	V	12

TABLE IX. Output resistance check test.

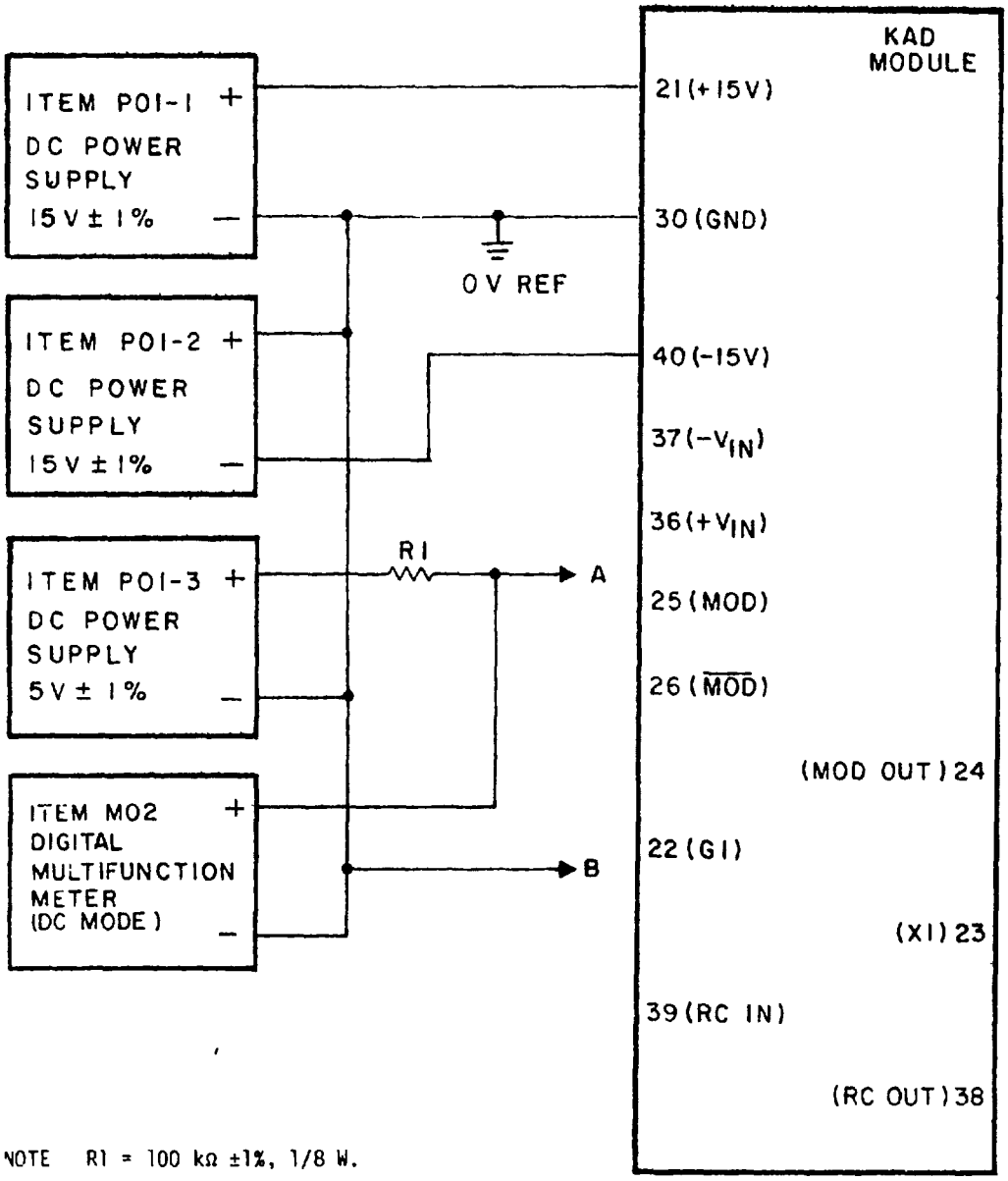
Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
H1	Output resistance check	3.53	5.05	V	13
H2	Output resistance check	-2.75	-5.05	V	13

TABLE X. Low pass filter network (60 Hz rejection) check test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
J	Low pass filter network 1/ (60 Hz rejection) check	80	117	mVrms	14

1/ Table I limits = 20 log (table X limits).

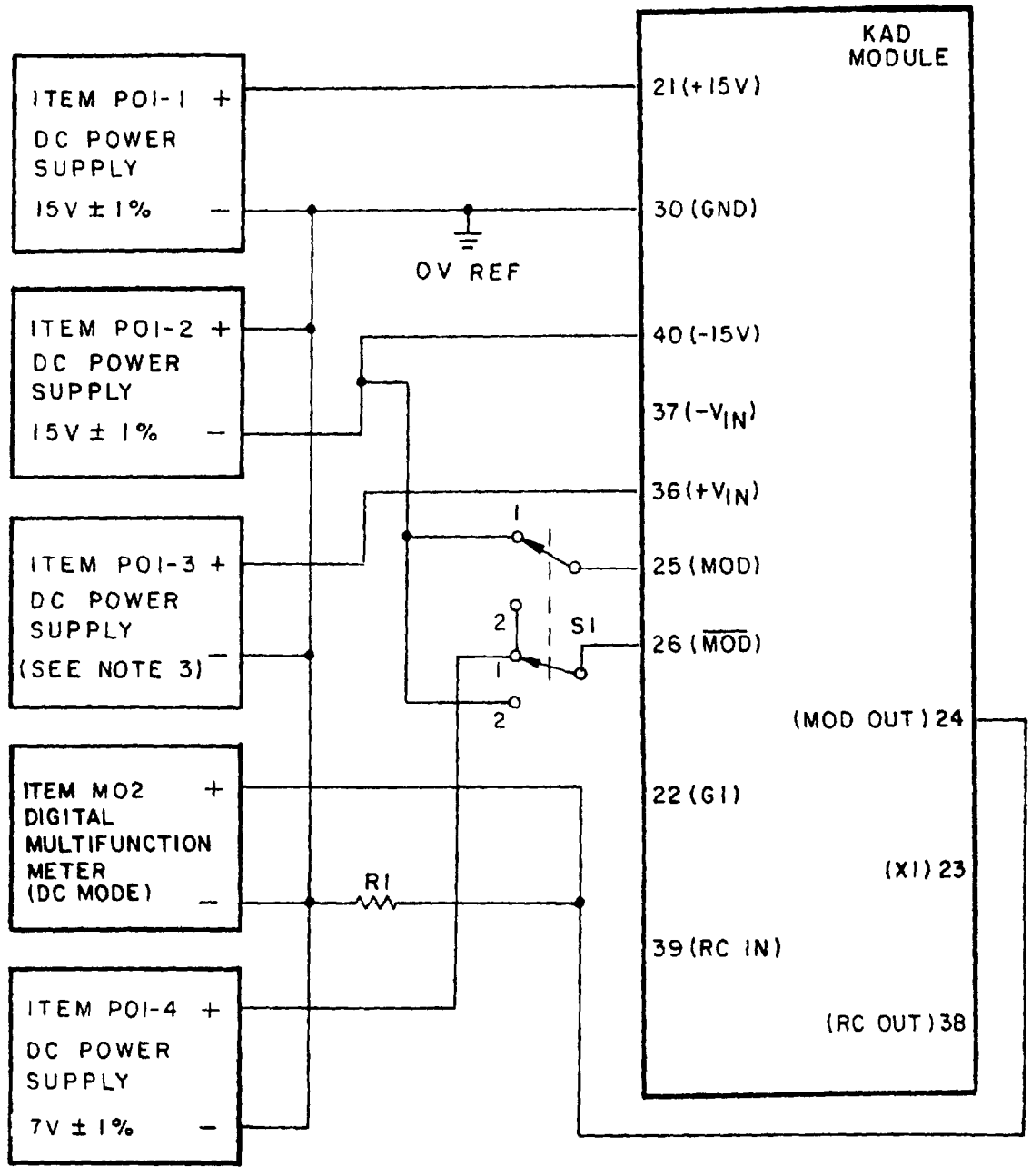
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NOTE R1 = 100 kΩ ±1%, 1/8 W.

FIGURE 12. Input resistance check test circuit for table VIII

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NOTES
1. R1 = 1 k ±1%, 1/8W
2. S1 = DPDT switch
3. Prior to making the tests disconnect resistor R1 from pin 24 and adjust item PO1-3 to +5 V ±1% for test condition H1 and -5 V ±1% for test condition H2
4. Set switch S1 to position 1 for test condition H1 and position 2 for test condition H2.

FIGURE 11. Output resistance check test circuit for table IV

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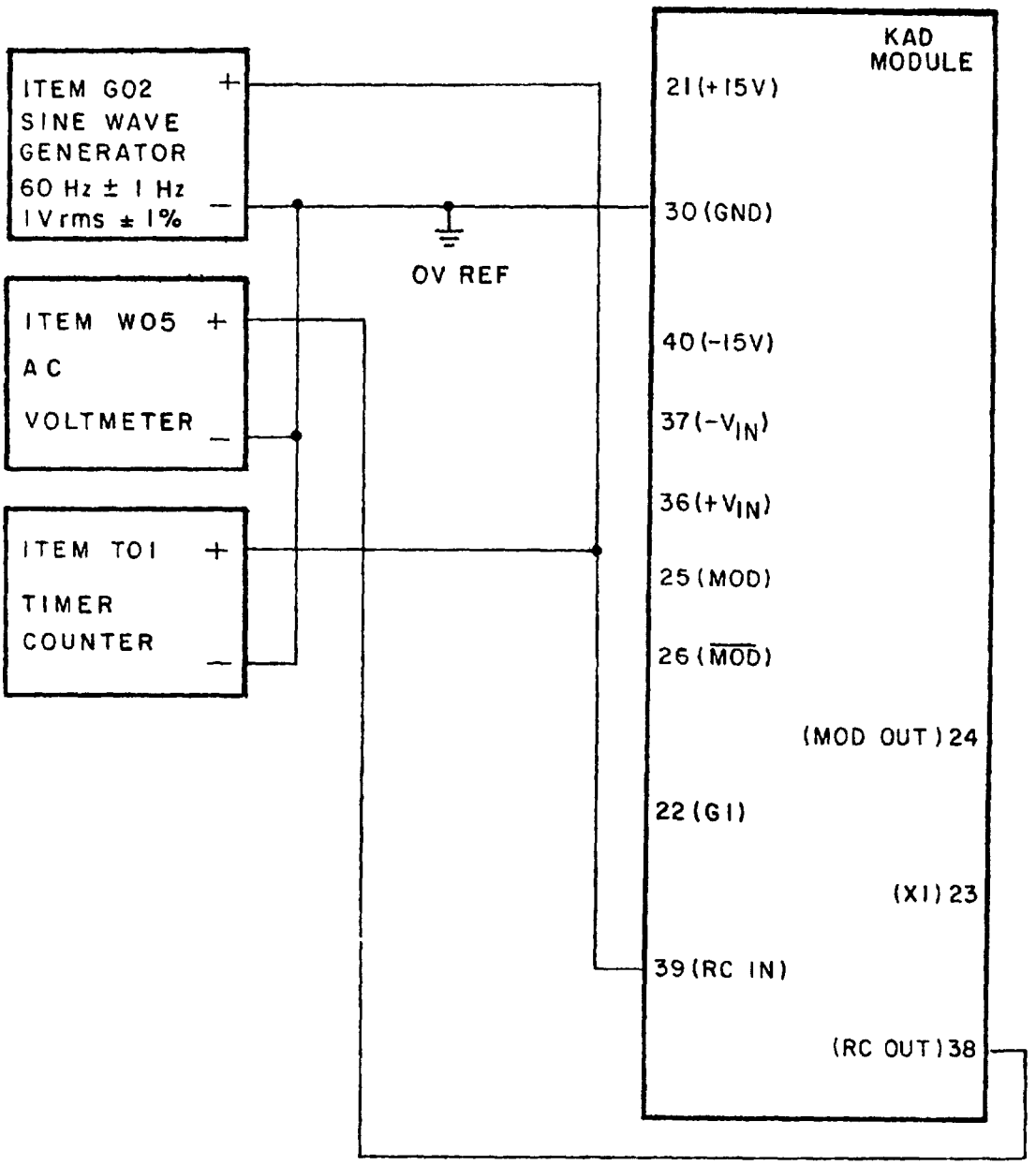


FIGURE 14. Low pass filter network (60 Hz rejection) check test circuit for table X

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TABLE XI. Output offset voltage test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
K1	Output offset voltage	-15	+15	mV	15
K2	Output offset voltage	-30	+30	mV	15

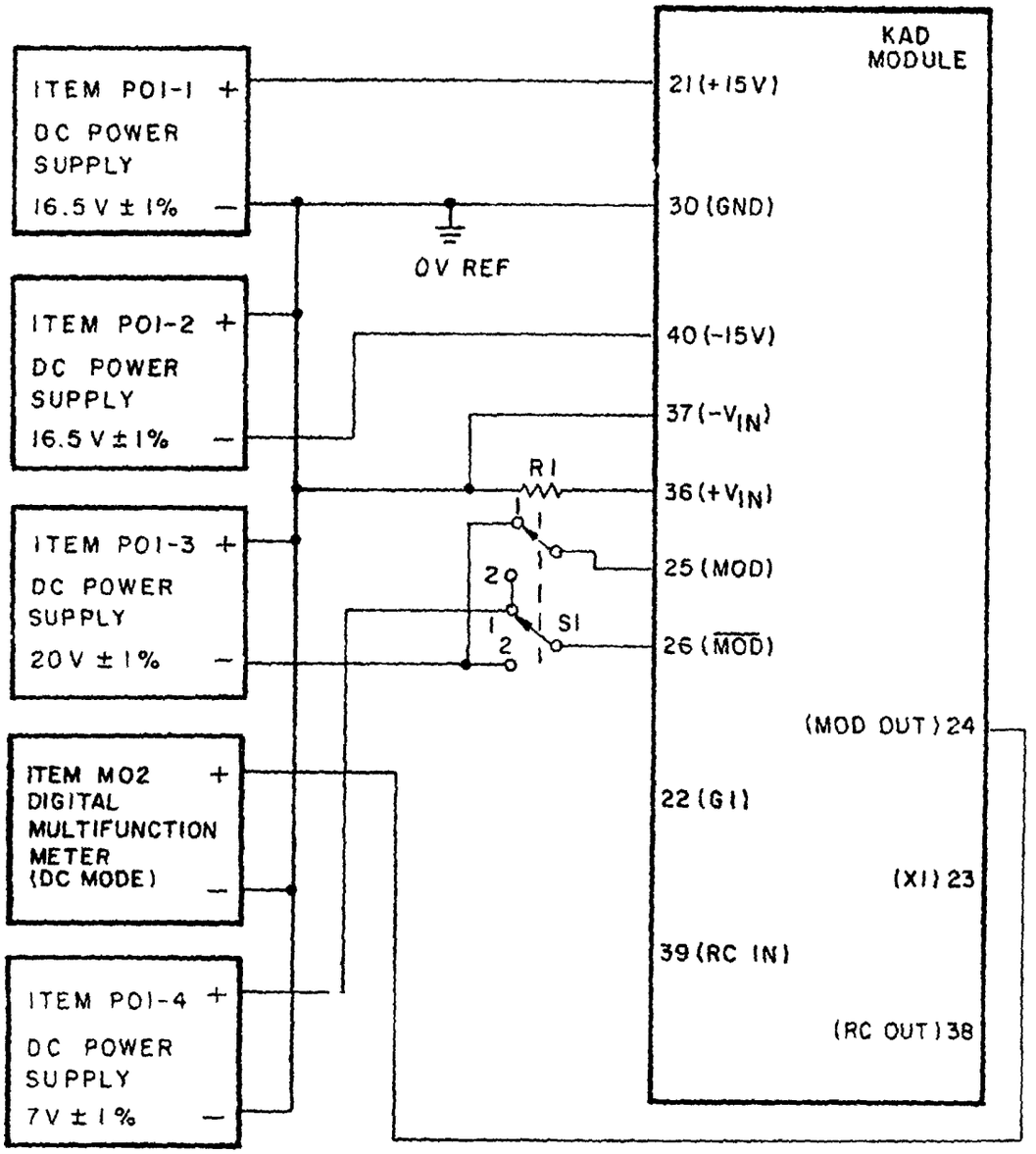
TABLE XII. FET ON resistance check test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	FET ON resistance check (V_{IN} = -10 V), 620 Ω	30	C	0	-580	mV	16
L2	FET ON resistance check (V_{IN} = +10 V), 205 Ω	C	30	0	+200	mV	16

TABLE XIII. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
M1	Power supply current ($+V_{CC}$)	1	8	mA	17
M2	Power supply current ($-V_{CC}$)	1	8	mA	17

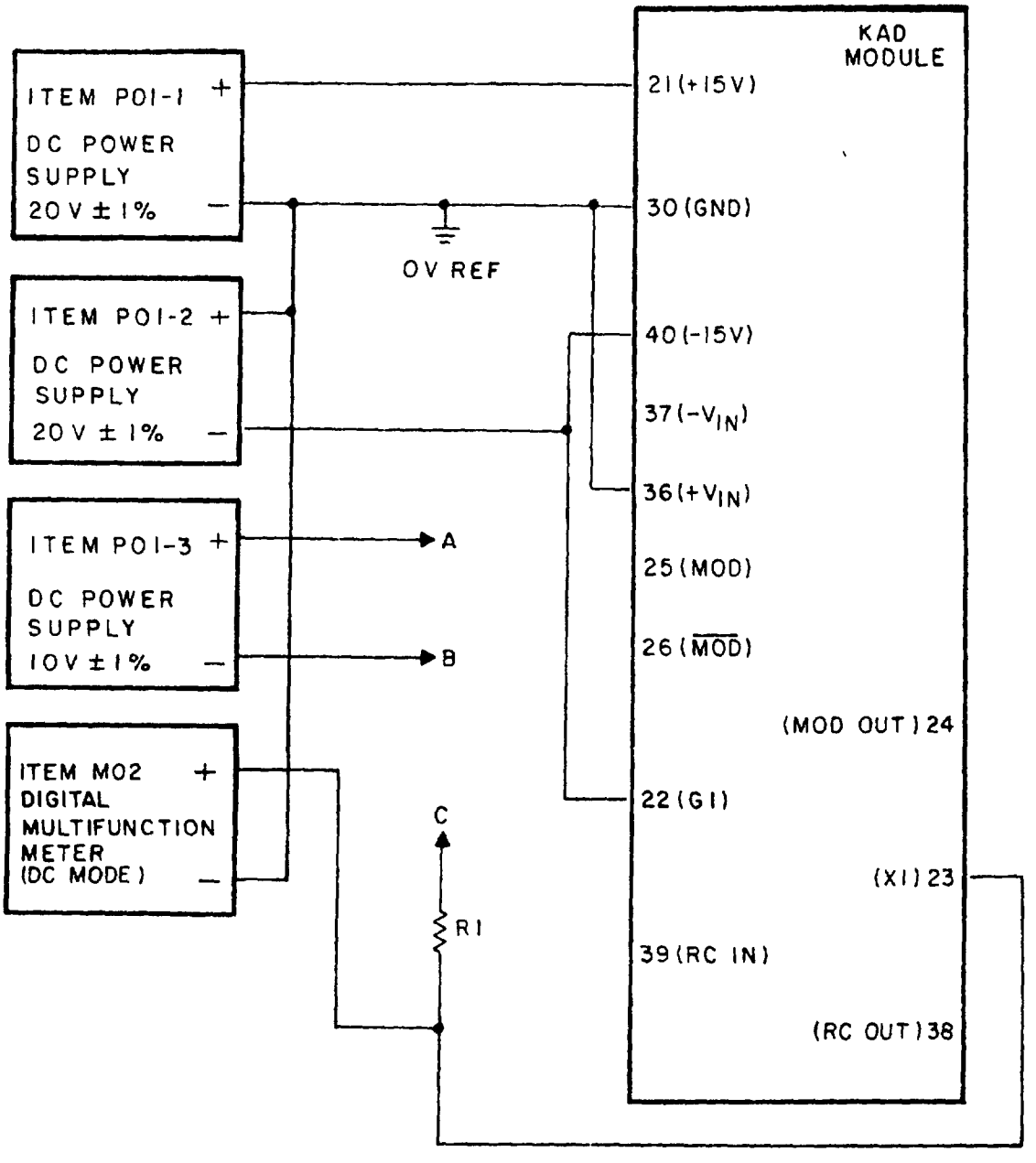
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NOTES
1 R1 = 28 kΩ ±1%, 1/8 W.
2 S1 = DPDT switch.
3 Set switch S1 to position 1 for test condition J1 and position 2 for test condition J2.

FIGURE 15 Output offset voltage circuit for table X1.

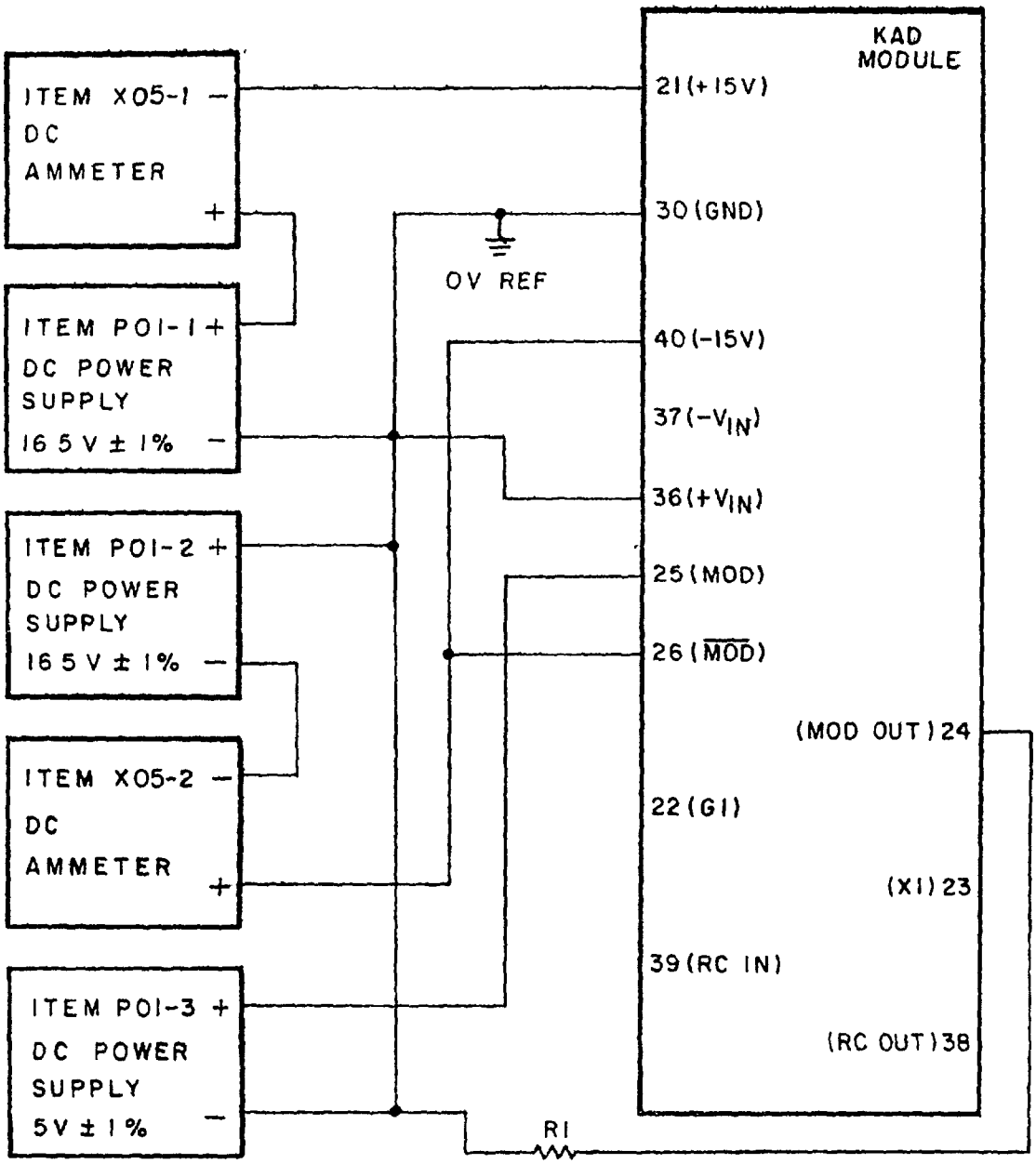
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NOTE R1 = 10 kΩ ±1 , 1/8 W

1/ R1 is [illegible] check test [illegible] table XII

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NOTE R1 = 10 kΩ ±1%, 1/8 W.

FIGURE 17. Power supply current test circuit for table XIII.

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TABLE XIV. Gate input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Gate input capacitance	22	30	0	15	pF	18
N2	Gate input capacitance	25	30	0	15	pF	18
N3	Gate input capacitance	26	30	0	15	pF	18

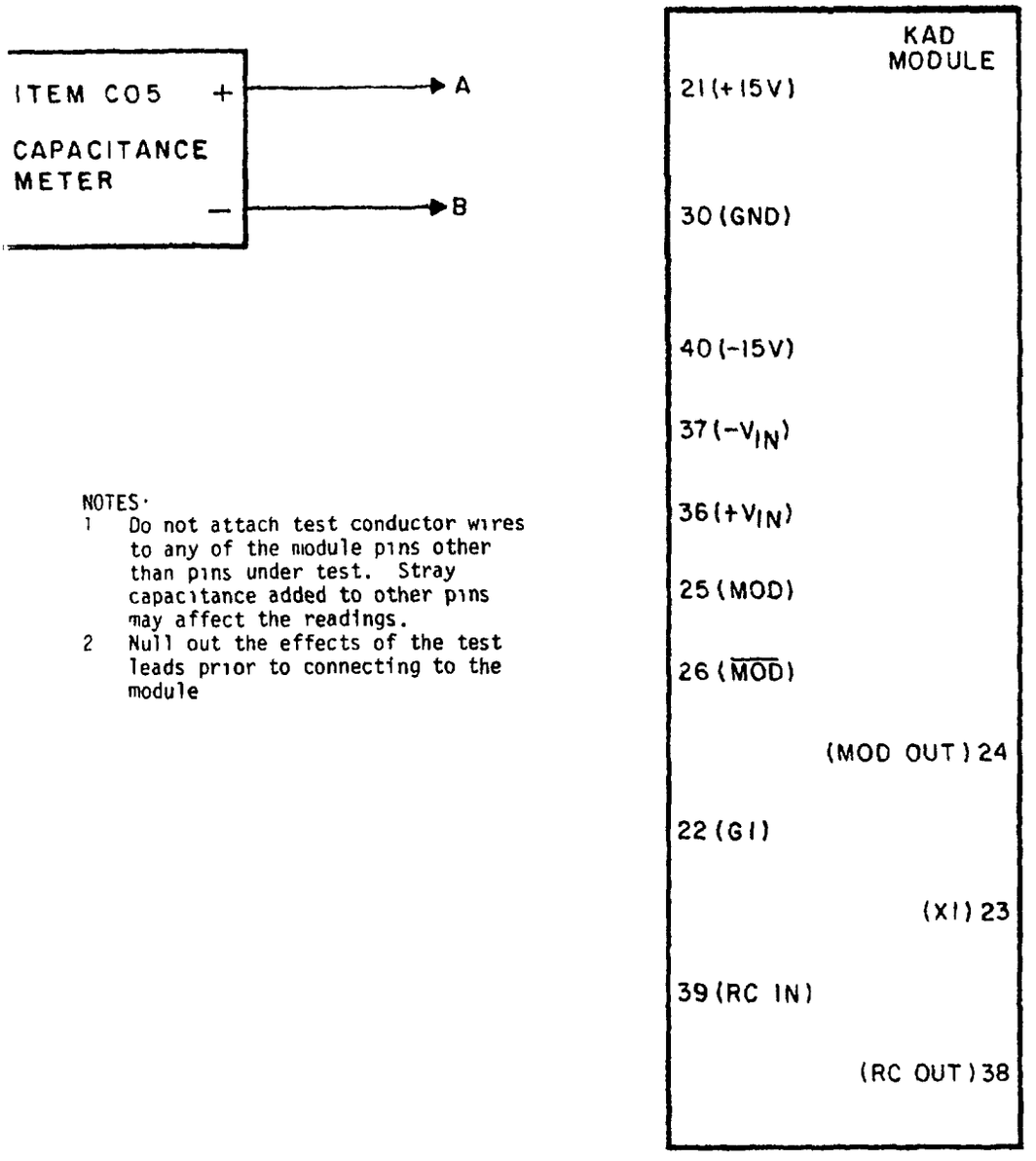
TABLE XV. Gate output capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Gate output capacitance	22	23	0	12	pF	18
P2	Gate output capacitance	25	24	0	12	pF	18
P3	Gate output capacitance	26	24	0	12	pF	18

TABLE XVI. Power supply (positive and negative) transient rise time test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		B to pin	Min	Max		
Q1	Power supply transient rise time, positive	21	25	75	µs	19,20
Q2	Power supply transient rise time, negative	40	25	75	µs	19,20

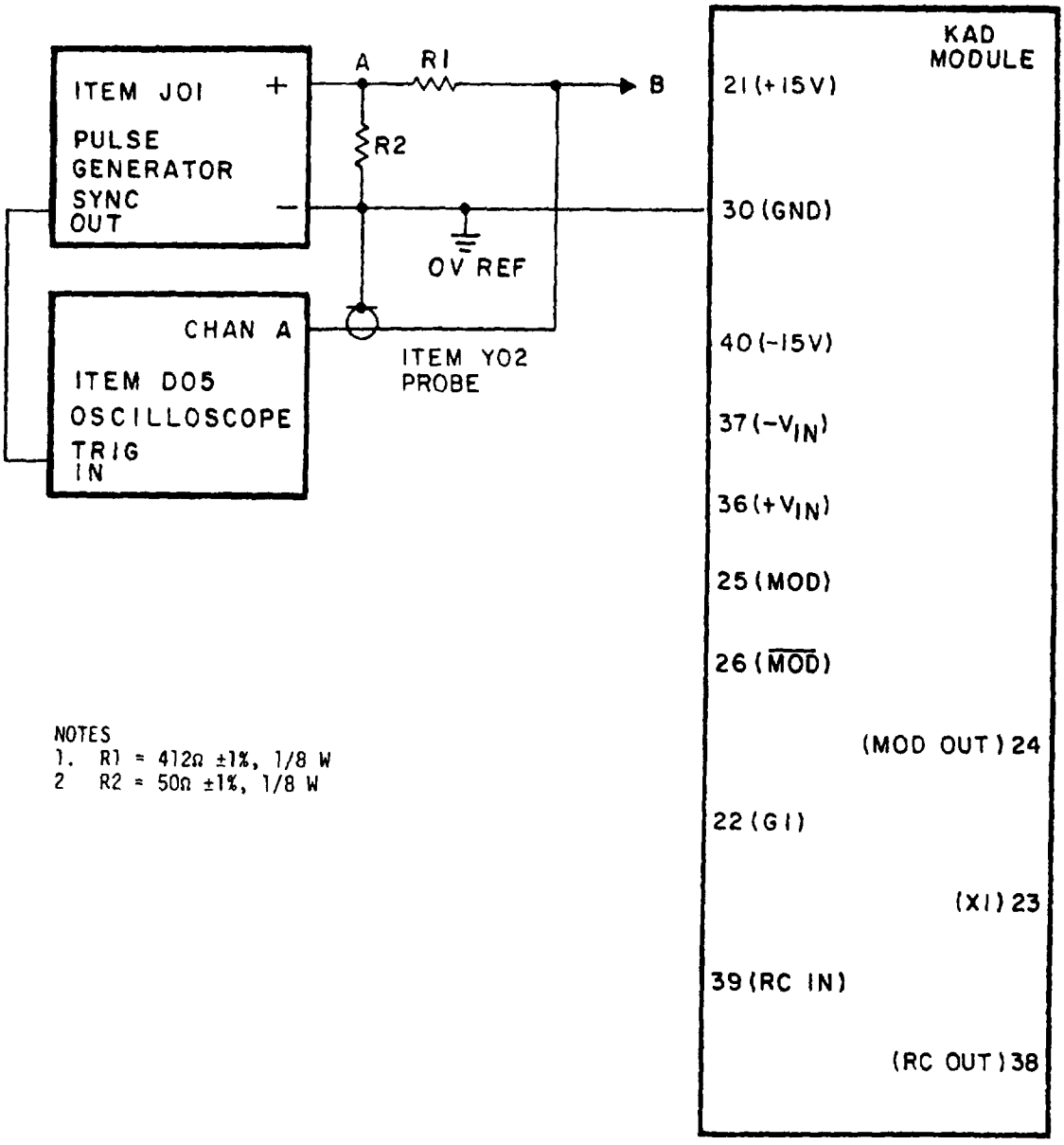
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NOTES:
1. Do not attach test conductor wires to any of the module pins other than pins under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting to the module

FIGURE 18. Gate (input and output) capacitance test circuit for tables XIV and XV

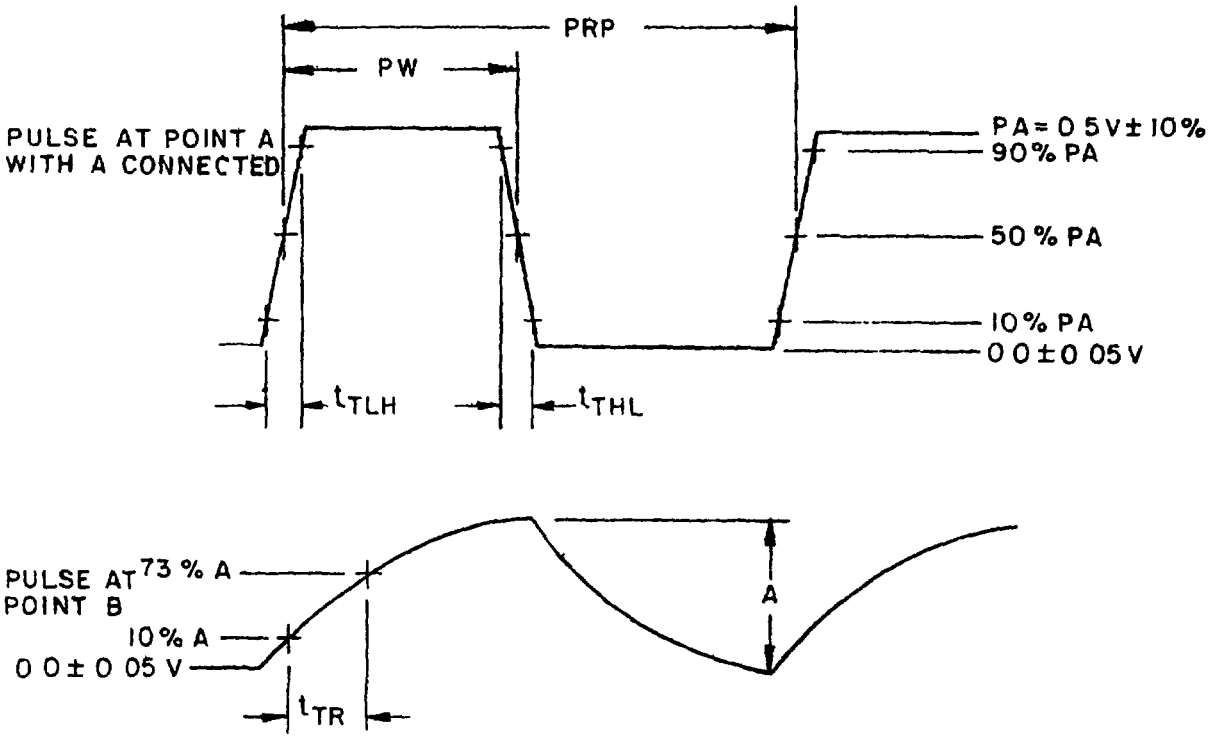
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NOTES
1. R1 = 412Ω ±1%, 1/8 W
2 R2 = 50Ω ±1%, 1/8 W

FIGURE 19 Power supply (positive and negative) transient rise time test circuit for table XVI

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NOTES
1 Input pulse parameters
PW = 300 µs ±10%
$t_{TLH} = t_{THL}$ = 0.50 µs maximum.
PRP = 600 µs ± 10% .
2 Measure transient rise time (t_{TR}) at point B.

FIGURE 20 Power supply (positive and negative) transient rise time test circuit waveforms for table XII

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